Prescriptions of corticosteroids in primary care and related clinical indications: a national population-based cohort study in England

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Summary

Background Corticosteroids are widely prescribed in primary care for a broad range of inflammatory and autoimmune conditions. Despite their therapeutic benefits, corticosteroids are associated with adverse effects, particularly with long-term use. We aimed to describe corticosteroid prescribing patterns to better understand the current landscape of corticosteroid use in England.

Methods We conducted a population-based cohort study using electronic health-care records from the Clinical Practice Research Datalink Aurum. The study population included adults (aged 18 years or older) who were alive, registered with a general practice in England, and had at least one corticosteroid prescription between Jan 1 and Dec 31, 2023 (the period of follow-up). The study population (age, sex, and ethnicity) and corticosteroid characteristics (drug substance and duration of use) were described by route of administration. Outcomes included pre-defined clinical indications across disease systems and were defined in the 2-year window prior to the patients' first corticosteroid prescription up until their last corticosteroid prescription and described by route of administration. Finally, combinations of disease systems were described. This study was registered with Clinical Practice Research Datalink (erap number 25_004991).

Findings Between Jan 1 and Dec 31, 2023, 12 029 952 unique corticosteroids were prescribed to 2 564 729 patients. Median age of patients was 56·5 years (IQR 40·5–70·5), 1 466 264 (57·2%) of the population were female, 1 999 601 (78·0%) were White, 207 228 (8·1%) were South Asian, 77 205 (3·0%) were Black, 33 239 (1·3%) were Mixed, 53 202 (2·1%) were of Other ethnicity, and 194 254 (7.6%) had missing ethnicity. The median number of corticosteroids prescribed per person over the year of follow-up was two (IQR 1–6). Inhaled corticosteroids accounted for the highest number of prescriptions (44·2%), followed by cutaneous (19·6%), nasal (14·3%), and oral (14·0%) routes. 1 267 685 (49·4%) patients had at least one pre-defined clinical indication recorded up to 2 years prior. Asthma (801 924 [63·3%]), eczema (229 382 [18·1%]), and chronic obstructive pulmonary disease (220 759 [17·4%]) were the most frequently recorded indications for all corticosteroids. Among patients with multiple indications, the most common combination was respiratory and dermatology-related disease with 53·8% of patients receiving corticosteroid treatment for both types of diseases.

Interpretation Corticosteroid prescribing remains widespread in primary care in England. These findings highlight the need to evaluate safer therapeutic alternatives and update current guidelines for potentially reducing corticosteroid exposures.

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Introduction

Corticosteroids are frequently used to manage symptoms and reduce disease progression in chronic and acute inflammatory and autoimmune diseases.[1] Corticosteroids have varied pharmacological effects, allow flexible administration, and are widely available. Specifically, corticosteroids are used to reduce inflammation and suppress the immune response, and are available in various formulations, including inhaled forms for respiratory conditions, topical applications for skin disorders, and oral tablets for a range of conditions.[2] Many chronic conditions require repeat corticosteroid prescriptions and long-term use to manage symptoms. Certain conditions might require the use of multiple types of corticosteroids to manage flare-ups in addition to ongoing treatment of the chronic disease.[3,4] Similarly, patients with multiple conditions might be prescribed different types of corticosteroids to manage each disease, potentially resulting in concurrent use of multiple corticosteroids.[5]

Corticosteroids are one of the most common drug groups that lead to adverse drug reactions in patients in hospital and in the community.[6] Accumulating evidence suggests that corticosteroids might be over-used or misused and can lead to steroid resistance, as well as potentially

Research in context

Evidence before this study

We searched PubMed from database inception for all previous literature on corticosteroid use in the UK, up to July 1, 2025. No language or start date restrictions were applied. Specifically, we searched for any studies with titles including ("corticosteroid*" or "glucocorticoid*") and ("use", "presrib*", or "prescription*") and the terms ("United Kingdom", "UK" or "England") in titles and abstracts. A total of 45 studies were identified: 13 studies investigated risk of adverse events or poor health outcomes due to corticosteroids in specific disease populations, 27 studies investigated use of corticosteroids in specific disease populations (eg, asthma), and one study was methodological. The remaining four studies described the use of corticosteroids in the UK across various conditions; however, these specifically focused on oral corticosteroids alone. No studies have focused on all possible routes of administration of corticosteroids.

Added value of this study

To our knowledge, this is the first study to describe corticosteroid prescriptions in the general population and determine which disease systems, routes of administration, and clinical indications were the most common in England. These data form a strong baseline for future work to evaluate safer therapeutic alternatives and reduce the burden of corticosteroids.

Implications of all the available evidence

Corticosteroid prescribing remains common in primary care, with 16·1% of the English population receiving at least one prescription in 2023. Although respiratory, dermatology, rheumatology, and gastrointestinal conditions were the most common disease systems for which corticosteroids were prescribed, the co-occurrence of conditions requiring corticosteroids also highlights the importance of optimal management and prescribing in patients with multi-organ disease. With the rise of corticosteroid-sparing treatments, further evaluation of their use, as well as the safety and cost-effectiveness of corticosteroids, is required to inform clinical guidelines for the management of people using corticosteroids.

life-threatening withdrawal reactions.[7–12] With increasing numbers of corticosteroid-sparing treatments now available for specific conditions, up-to-date evidence about the utilisation of corticosteroids is needed.

Using routinely collected electronic health-care records from primary care, we aimed to describe recent corticosteroid prescribing patterns over a year in England and determine common clinical indications and combinations of disease systems requiring corticosteroids, and common routes of administration.

Methods

Study design and participants

Routinely collected electronic health records from Clinical Practice Research Datalink (CPRD) Aurum were used (CPRD protocol number: 25_004991) to obtain data for this national, population-based cohort study. CPRD Aurum is a primary care electronic health records database covering approximately 20% of primary care general practices in England that have consented to contributing data. CPRD Aurum is representative of the population of England in terms of geographical spread, socioeconomic deprivation, age, and sex.[13–15] Primary care data from CPRD-contributing practices are collected by CPRD every month, added to previous data, and are available for download from CPRD.

For the **CPRD online summary** see https://www.cprd.com/approved-studies/clinical-indications-and-prescribing-patterns-corticosteroids-england

The study population included patients who were: (1) registered at a general practice before Jan 1, 2023; (2) aged 18 years or older on Jan 1, 2023; (3) had at least one corticosteroid prescription between Jan 1 and Dec 31, 2023; and (4) were alive and registered at a general practice between Jan 1 and Dec 31, 2023. Jan 1, 2023, was defined as the index date to ensure up-to-date data collection of corticosteroid prescription patterns. End of follow-up was defined as Dec 31, 2023. No exclusion criteria were applied based on comorbidities. Only adults were included in this study due to differences in prescribing patterns and guidelines between adults and children for a range of clinical indications.

The study was conducted with the approval of the UK's Health Research Authority Research Ethics Committee (CPRD erap number 25_004991). Given the observational nature of the study and the use of anonymised data provided by the CPRD, no additional ethical approval or consent was required. The design and conduct of the study adhered to the regulations for public health research in the UK, ensuring compliance with CPRD data governance standards. This study is reported in line with the RECORD Statement guidelines. This study was registered with CPRD (erap number 25_004991) and an online summary of the study has been published.

Procedures

Demographic data were defined using CPRD data. These included age at index date, sex, and ethnicity. Age at index date was calculated by subtracting the recorded date of birth from the index date. Sex was defined based on self-reported sex when patients registered with their general practice and was defined as male or female. Ethnicity was recorded as White, Black, South Asian, Mixed, Other, and missing ethnicity based on previously published codes.[15]

Corticosteroid prescription data were defined between Jan 1 and Dec 31, 2023, in CPRD Aurum data. Corticosteroid prescriptions were defined as any corticosteroid prescription for any disease system. Corticosteroid code lists were created by searching CPRD's product dictionary for corticosteroid-related medications and were reviewed by clinical experts (MJ, FAP, SB, and JKQ) to ensure bias in the classification of corticosteroids was reduced.[16] Additional prescription-related information included route of administration, drug substance, and length of time that corticosteroids were prescribed. Route of administration was defined by CPRD and included cutaneous, gastroenteral, infiltration, intraarticular, intrabursal, intradermal, intralesional, intramuscular, intravenous, intravitreal, nasal, ocular, oral, auricular, oromucosal, periarticular, rectal, subconjunctival, and subcutaneous. Drug substance was defined in CPRD Aurum and included the active ingredient of the prescribed corticosteroid. Length of time that corticosteroids were prescribed was defined as 3 months of the year of follow-up, 6 months of the year, 9 months of the year, and the full 12 months of follow-up. Patients were required to have at least one corticosteroid prescription in a given quarter of a year to be classified into each group. Strength of drug substance was based on dose, potency, and route of administration and manually categorised as low, medium, and high based on the dose, potency, and route of administration (appendix pp 7–10).

Disease systems wherein corticosteroids are commonly prescribed were collated a priori through discussions with clinical experts. Disease systems included dermatology, endocrine, gastrointestinal, neurology, renal, respiratory, rheumatology, and ophthalmology. Specific clinical indications within each disease system were determined through previous work and discussion with specialist clinicians (MJ, FAP, SB, JKQ) and are summarised in the panel.[17] For each clinical indication, code lists were generated and reviewed by clinical experts (appendix p 2). These code lists were applied to the CPRD data to identify clinical indications within a window of 2 years prior to the first corticosteroid prescription up until the latest corticosteroid prescription in the follow-up year.

Statistical analysis

Descriptive statistics were used to summarise route of corticosteroid administration across the study population. Demographic and clinical characteristics were reported for the overall cohort (using proportions and medians [IQR]) and stratified by patients prescribed at least one corticosteroid for a specific route of administration. Stratification was conducted for six of the most common routes.

The proportions of patients with no recorded clinical indication, any indication, and multiple distinct indications were described using descriptive statistics and stratified by age and sex. Median (IQR) number of corticosteroid prescriptions over follow-up was described for the total population. The proportion of each specific clinical indication was reported for the entire study population and stratified by patients prescribed at least one corticosteroid for a specific route of administration. The proportions of people with specific clinical indications by respiratory, rheumatology, dermatology, and gastrointestinal disease systems were calculated. Similarly, proportions of people with a

Panel: Clinical indications for steroid use

Respiratory
- Asthma
- Chronic obstructive pulmonary disease
- Pulmonary fibrosis
- Bronchopulmonary aspergillosis
- Nasal polyps
- Chronic sinusitis
- Allergic rhinitis

Rheumatology
- Crystal arthropathies
- Rheumatoid arthritis
- Polymyalgia rheumatica
- Psoriatic arthritis
- Systemic lupus erythematosus
- Ankylosing spondylitis
- Idiopathic inflammatory myositis
- Giant cell arteritis
- Stills disease
- Small and medium vasculitis
- Other large vessel vasculitis
- Behcet's disease
- Systemic sclerosis
- IgG4 disease

Renal
- Membranous nephropathy
- Nephrotic syndrome
- IgA nephropathy
- Focal segmental glomerulonephritis
- Minimum change glom
- Acute interstitial nephritis
- Post kidney transplant

Endocrine
- Subacute thyroiditis
- Hyperaldosteronism
- Graves ophthalmopathy
- Adrenal insufficiency
- Congenital adrenal hyperplasia
- Hypopituitarism

Dermatology
- Eczema
- Vitiligo
- Stevens-Johnson's syndrome
- Pyoderma gangrenosum
- Lichen planus
- Blistering conditions
- Alopecia

Gastrointestinal
- Inflammatory bowel disease
- Oral lichen planus
- Eosinophilic oesophagitis
- Autoimmune hepatitis

(Continues on next page)

See Online for appendix

Panel (continued from previous page)

Ophthalmology
- Uveitis (non-infectious)

Neurological
- Myasthenia gravis
- Multiple sclerosis
- Chronic inflammatory demyelinating polyneuropathy
- Bell's Palsy

Additional search terms and synonyms for clinical indications listed are available in the appendix (p 2).

specific route of corticosteroid prescription for each clinical indication were calculated.

As a sensitivity analysis, clinical indications were defined within a 5-year window prior to the first corticosteroid prescription following previous work evaluating clinical indications in CPRD, which found that codes for chronic conditions might not be entered repeatedly.[18] Similarly, clinical indications 2 years before the first corticosteroid prescription were defined in patients who had been registered with their current general practice for at least 1 year before Jan 1, 2023, as an additional sensitivity analysis. Additionally, analyses describing clinical indications 2 years prior to the first corticosteroid prescription were repeated by sex to determine whether differences in clinical indications within 2 years differed between males and females.

For patients with clinical indications for two different disease systems, the proportion of each possible combination of disease systems was described. For all descriptive analyses, counts fewer than ten, and any associated potentially identifiable values, were suppressed in accordance with disclosure control policies.[19] Missing data were not imputed and were reported when necessary. All data were presented as absolute numbers and proportions. All data management and analyses were done with STATA (version 18.5).

Role of the funding source

There was no funding source for this study.

Results

Between Jan 1 and Dec 31, 2023, a total of 12 029 952 individual corticosteroid prescriptions were made in 2 564 729 patients (appendix p 22). This corresponded to 16·1% of the CPRD population in 2023 (total population with follow-up between Jan 1, 2023, and Dec 31, 2023, was 15 953 200). The median number of corticosteroids prescribed per person over the year of follow-up was two (IQR 1–6). The most common route of administration was inhalation with 5 321 403 (44·2%) corticosteroid prescriptions, followed by cutaneous (2 358 070 [19·6%]), nasal (1 722 835 [14·3%]), oral (1 683 256 [14·0%]), auricular (309 267 [2·6%]), ocular (151 043 [1·3%]), and rectal or cutaneous (147 509 [1·2%]; figure 1). All other routes of administration were less than 1% of the total number of corticosteroid prescriptions (appendix p 4). Despite the largest proportion of prescriptions corresponding to the inhalation route of administration, the largest proportion of patients were prescribed corticosteroids for cutaneous routes (1 006 463 [39·2%] of 2 564 729; figure 1).

Overall, the median age of the study population was 56·5 years (IQR 40·5–70·5). Individuals with at least one ocular corticosteroid prescription were older (median age 72·5 years [IQR 60·5–79·5]) and individuals with at least one rectal or cutaneous corticosteroid prescription were younger (median age 50·5 years [IQR 36·5–67·5]) than individuals with other routes of corticosteroids. The proportion of females prescribed at least one corticosteroid was higher than males (1 466 264 [57·2%] *vs* 1 098 465 [42·8%]) and ranged from 56·3% to 61·0%, depending on the route of administration. In terms of ethnicity, 2 370 475 (92·4%) of the study population had ethnicity recorded and most of the population had White ethnicity (1 999 601 [78·0%]), followed by South Asian (207 228 [8·1%]) and Black ethnicities (77 205 [3·0%]; appendix p 5).

The most common drug substances for inhaled corticosteroids included beclometasone (635 719 [69·6%]), followed by fluticasone (177 698 [19·4%]) and budesonide (152 940 [16·7%]; appendix p 5). Dosages for drug substances by each route of administration, as well as their strength based on a combination of dose, potency, and route of administration, are presented in the appendix (p 7). For inhaled corticosteroids, 791 009 (86·5%) of 913 733 patients were prescribed low strength corticosteroids and 178 343 (19·5%) were prescribed medium strength corticosteroids. For cutaneous corticosteroids, the most common drug substance included hydrocortisone (442 160 [43·9%]) and betamethasone (435 790 [43·3%]), followed by clobestasone (163 043 [16·2%]; appendix p 5). A total of 515 399 (51·2%) of 1 006 463 patients were prescribed low strength corticosteroids, 173 692 (17·3%) were prescribed medium strength corticosteroids, and 99 787 (9·9%) were prescribed high strength corticosteroids. For nasal corticosteroids, the most common drug substance was mometasone (260 092 [47·6%]), fluticasone (218 646 [40·0%]), and beclomethasone (110 726 [20·3%]; appendix p 5). A total of 374 404 (68·5%) of 546 570 patients were prescribed low strength corticosteroids, 218 646 (40·0%) were prescribed medium strength corticosteroids, and 1957 (0·4%) were prescribed high strength corticosteroids. For oral corticosteroids, the most common drug substance was prednisolone (408 535 [92·7%]), fludrocortisone (11 703 [2·7%]), and betamethasone (6532 [1·5%]; appendix p 5). A total of 11 752 (2·7%) of 440 775 patients were prescribed low strength corticosteroids, 430 500 (97·7%) were prescribed medium strength corticosteroids, and 6391 (1·4%) were prescribed high strength corticosteroids. For ocular corticosteroids, the most common drug substance was dexamethasone (33 381 [70·6%]) and prednisolone (10 850 [23·0%]; appendix p 5). All ocular-related corticosteroids were low strength. Finally, rectal or cutaneous corticosteroids included hydrocortisone (49 960 [52·1%]) and prednisolone (49 387 [51·5%]) and all prescriptions were medium strength corticosteroids (appendix pp 5–7).

Of individuals prescribed inhaled corticosteroids, 438 953 (48%) were prescribed corticosteroids for the whole year of follow-up. For all other routes of administration, the most common length of time for which corticosteroids were prescribed was a quarter of a year (cutaneous, 68·8%; nasal, 55·2%; oral, 59·2%; ocular, 62·2%; rectal or cutaneous, 84·8%; appendix p 5).

In the total study population, 1 267 685 (49·4%) patients had at least one clinical indication recorded within the 2-year window of interest (table). The proportion of people with at least one clinical indication in the population who had at least 1 year of registration at a general practice was similar (1 192 386 [49·6%] of 2 406 076), as was the proportion of people with a clinical indication within 5 years (1 297 666 [50·6%] of 2 564 729), and these proportions were also similar by sex (appendix pp 11–13, 15). Of those who had a clinical indication within 2 years, 1 002 052 (79·0%) had one clinical indication recorded, 230 727 (18·2%) had two clinical indications, 31 161 (2·5%) had three clinical indications, and 3745 (0·3%) had four or more clinical indications recorded (table). Additionally, females had a higher proportion of clinical indications than did males. Of the 1 297 044 patients who had no clinical indications recorded, 734 519 (56·6%) were female, whereas of the 3745 who had four or more clinical indications recorded, 2288 (61·1%) were female. The median age of individuals increased with increasing number of clinical indications and was 56·5 years (IQR 39·5–70·5) in those with no clinical indications recorded, compared with 67·5 years (56·7–76·5) in those with four or more clinical indications reported.

In the total study population, asthma was the most common clinical indication (801 924 [63·3%] of 1 267 685), followed by eczema (229 382 [18·1%]) and chronic obstructive pulmonary disease (COPD; 220 759 [17·4%]). Other relatively common clinical indications included allergic rhinitis (57 294 [4·5%]), rheumatoid arthritis (45 099 [3·6%]), crystal arthropathies (42 186 [3·3%]), polymyalgia rheumatica (26 648 [2·1%]), alopecia (26 620 [2·1%]), and inflammatory bowel disease (22 037 [1·7%]; table).

Of the 913 733 individuals prescribed inhaled corticosteroids, 825 417 (90·3%) had a clinical indication recorded within 2 years of their latest corticosteroid prescription. Of those with at least one clinical indication, the most common indications included asthma (720 613 [87·3%]) and COPD (176 535 [21·4%]; table). Of those prescribed cutaneous corticosteroids, 388 125 (38·6%) had a clinical indication recorded within 2 years. Of those with at least one clinical indication, the most common indications included eczema (184 224 [47·5%]) and alopecia (14 918 [3·8%]; table; appendix p 23). Of those prescribed nasal steroids, 222 067 (40·6%) had at least one clinical indication within 2 years. Of those with at least one, the most common indications included asthma (131 752 [59·3%]) and COPD (30 745 [13·8%]; table). Of those prescribed oral corticosteroids, 316 776 (71·9%) had a clinical indication recorded within 2 years. Of those with at least one indication, the most common included asthma (180 894 [57·2%]), COPD (108 380 [34·3%]), eczema (25 272 [8·0%]), polymyalgia rheumatica (23 509 [7·4%]), rheumatoid arthritis (19 290 [6·1%]), crystal arthropathies (13 008 [4·1%]), and alopecia (3222 [1·0%]; table). Of those prescribed ocular corticosteroids, 16 259 (34·4%) had a clinical indication recorded within 2 years. Of those with at least one, 3461 (21·3%) had a clinical indication for uveitis (table). Finally, of those prescribed rectal or cutaneous corticosteroids, 24 174 (25·2%) had a clinical indication recorded within 2 years. Of those with at least one indication, 4569 (18·9%) had a clinical indication recorded for eczema (table). Notably, due to the nature of the study design, other clinical indications might have been recorded that were not always related to a specific corticosteroid prescription.

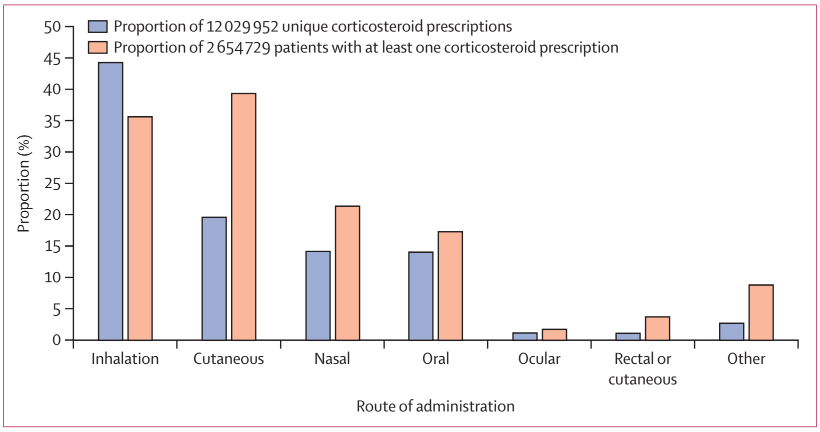

***Figure 1:* Frequency of route of administration of corticosteroid prescriptions in 2023**
Although estimates for the blue bars are mutually exclusive, estimates for the orange bars are not mutually exclude as patients could have had corticosteroids prescribed for more than one type or route of administration. The column "Other" includes all other routes of administration for which less than 1% of corticosteroid prescriptions were prescribed for a particular route of administration. Further details on absolute numbers and proportions are provided in the appendix (p 4).

The most common clinical indications recorded for the four most common disease systems—respiratory, rheumatology, dermatology, and gastrointestinal—are presented in figure 2 and the appendix (p 18). The most common route of administration for asthma, COPD, bronchopulmonary aspergillosis, hyperaldosteronism, graves ophthalmopathy, and eosinophilic oesophagitis was inhalation. Clinical indications where cutaneous routes of administration were the most common included crystal arthropathies, psoriatic arthritis, ankylosing spondylitis, IgA nephropathy, subacute thyroiditis, multiple sclerosis, and all cutaneous-related indications. Nasal polyps, chronic sinusitis, and allergic rhinitis were indications where the nasal route of administration was the most common. The ocular route of administration was the most common for uveitis, and all other clinical indications were most prescribed via the oral route of administration (appendix p 18).

	All patients (N=2 564 729)	Patients prescribed inhaled corticosteroids (N=913 733)	Patients prescribed cutaneous corticosteroids (N=1 006 463)	Patients prescribed nasal corticosteroids (N=546 570)	Patients prescribed oral corticosteroids (N=440 775)	Patients prescribed ocular corticosteroids (N=47 249)	Patients prescribed rectal or cutaneous corticosteroids (N=95 971)
No indication	1 297 044 (50·6%)	88 316 (9·7%)	618 338 (61·4%)	324 503 (59·4%)	123 999 (28·1%)	30 990 (65·6%)	71 797 (74·8%)
Any indication	1 267 685 (49·4%)	825 417 (90·3%)	388 125 (38·6%)	222 067 (40·6%)	316 776 (71·9%)	16 259 (34·4%)	24 174 (25·2%)
Number of clinical indications							
1	1 002 052/1 267 685 (79·0%)	629 182/825 417 (76·2%)	298 751/388 125 (77·0%)	169 660/222 067 (76·4%)	222 744/316 776 (70·3%)	12 212/16 259 (75·1%)	19 643/24 174 (81·3%)
2	230 727/1 267 685 (18·2%)	169 838/825 417 (20·6%)	76 416/388 125 (19·7%)	44 105/222 067 (19·9%)	79 211/316 776 (25·0%)	3356/16 259 (20·6%)	3913/24 174 (16·2%)
3	31 161/1 267 685 (2·5%)	23 598/825 417 (2·9%)	11 544/388 125 (3·0%)	7323/222 067 (3·3%)	12 921/316 776 (4·1%)	588/16 259 (3·6%)	551/24 174 (2·3%)
4 or more	3745/1 267 685 (0·3%)	2799/825 417 (0·3%)	1414/388 125 (0·4%)	979/222 067 (0·4%)	1900/316 776 (0·6%)	103/16 259 (0·6%)	67/24 174 (0·3%)
Respiratory							
Asthma	801 924/1 267 685 (63·3%)	720 613/825 417 (87·3%)	154 607/388 125 (39·8)	131 752/222 067 (59·3%)	180 894/316 776 (57·1%)	6571/16 259 (40·4%)	12 771/24 174 (52·8)
Chronic obstructive pulmonary disease	220 759/1 267 685 (17·4%)	176 535/825 417 (21·4%)	48 222/388 125 (12·4%)	30 745/222 067 (13·8%)	108 380/316 776 (34·2%)	3283/16 259 (20·2%)	3493/24 174 (14·4)
Pulmonary fibrosis	9775/1 267 685 (0·8%)	4367/825 417 (0·5%)	2770/388 125 (0·7%)	1812/222 067 (0·8%)	5109/316 776 (1·6%)	210/16 259 (1·3%)	203/24 174 (0·8%)
Bronchopulmonary aspergillosis	517/1 267 685 (<0·1%)	493/825 417 (0·1%)	91//388 125 (<0·1%)	146/222 067 (0·1%)	308/316 776 (0·1%)	<10/16 259 (<0·1%)	<10/24 174 (<0·1%)
Nasal polyps	1182/1 267 685 (0·1%)	627/825 417 (0·1%)	180/388 125 (<0·1%)	879/222 067 (0·4%)	254/316 776 (0·1%)	<10/16 259 (<0·1%)	<10/24 174 (<0·1%)
Chronic sinusitis	21 361/1 267 685 (1·7%)	6367/825 417 (0·8%)	3606/388 125 (0·9%)	17 421/222 067 (7·8%)	3284/316 776 (1·0%)	181/16 259 (1·1%)	441/24 174 (1·8%)
Allergic rhinitis	57 294/1 267 685 (4·5%)	21 976/825 417 (2·7%)	13 432/388 125 (3·5%)	34 732/222 067 (15·6%)	7546/316 776 (2·4%)	458/16 259 (2·8%)	1415/24 174 (5·9%)
Rheumatology							
Crystal arthropathies	42 186/1 267 685 (3·3%)	14 614/825 417 (1·8%)	15 279/388 125 (3·9%)	7631/222 067 (3·4%)	13 008/316 776 (4·1%)	823/16 259 (5·1%)	1160/24 174 (4·8%)
Rheumatoid arthritis	45 099/1 267 685 (3·6%)	16 136/825 417 (2·0%)	14 792/388 125 (3·8%)	8469//222 067 (3·8%)	19 290/316 776 (6·1%)	1298/16 259 (8·0%)	1225/24 174 (5·1%)
Polymyalgia rheumatica	26 648/1 267 685 (2·1%)	4720/825 417 (0·6%)	4746/388 125 (1·2%)	2721/222 067 (1·2%)	23 509/316 776 (7·4%)	445/16 259 (2·7%)	400/24 174 (1·7%)
Psoriatic arthritis	8406/1 267 685 (0·7%)	1962/825 417 (0·2%)	5554/388 125 (1·4%)	1200/222 067 (0·5%)	1820/316 776 (0·6%)	126/16 259 (0·8%)	168/24 174 (0·7%)
Systematic lupus erythematosus	3528/1 267 685 (0·3%)	830/825 417 (0·1%)	1435/388 125 (0·4%)	585/222 067 (0·3%)	1897/316 776 (0·6%)	62/16 259 (0·4%)	111/24 174 (0·5%)
Ankylosing spondylitis	3361/1 267 685 (0·3%)	1023/825 417 (0·1%)	1334/388 125 (0·3%)	758/222 067 (0·3%)	719/316 776 (0·2%)	242/16 259 (1·5%)	122/24 174 (0·5%)
Idiopathic inflammatory myositis	629/1 267 685 (<0·1%)	130/825 417 (<0·1%)	207/388 125 (0·1%)	89/222 067 (<0·1%)	420/316 776 (0·1%)	11/16 259 (0·1%)	16/24 174 (0·1%)
Giant cell arteritis	4216/1 267 685 (0·3%)	1012/825 417 (0·1%)	880/388 125 (0·2%)	500/222 067 (0·2%)	3477/316 776 (1·1%)	91/16 259 (0·6%)	84/24 174 (0·3%)
Stills disease	146/1 267 685 (<0·1%)	21/825 417 (<0·1%)	40/388 125 (<0·1%)	18/222 067 (<0·1%)	96/316 776 (<0·1%)	<10/16 259 (<0·1%)	<10/24 174 (<0·1%)
Small and medium vasculitis	335/1 267 685 (<0·1%)	74/825 417 (<0·1%)	109/388 125 (<0·1%)	42/222 067 (<0·1%)	172/316 776 (0·1%)	42/16 259 (0·3%)	10/24 174 (<0·1%)
Other large vessel vasculitis	92/1 267 685 (<0·1%)	<10/825 417 (<0·1%)	16/388 125 (<0·1%)	<10/222 067 (<0·1%)	75/316 776 (<0·1%)	<10/16 259 (<0·1%)	<10/24 174 (<0·1%)
Bechet's disease	434/1 267 685 (<0·1%)	104/825 417 (<0·1%)	148/388 125 (<0·1%)	60/222 067 (<0·1%)	276/316 776 (0·1%)	24/16 259 (0·1%)	20/24 174 (0·1%)
Systemic sclerosis	682/1 267 685 (0·1%)	191/825 417 (<0·1%)	260/388 125 (0·1%)	135/222 067 (0·1%)	276/316 776 (0·1%)	<10/16 259 (<0·1%)	20/24 174 (0·1%)
IgG4 disease	<10/1 267 685 (<0·1%)	<10/825 417 (<0·1%)	<15/388 125 (<0·1%)	<10/222 067 (<0·1%)	<35/316 776 (<0·1%)	<10/16 259 (<0·1%)	<10/24 174 (<0·1%)
Renal							
Membranous nephropathy	188/1 267 685 (<0·1%)	63/825 417 (<0·1%)	63/388 125 (<0·1%)	33/222 067 (<0·1%)	87/316 776 (<0·1%)	<10/16 259 (<0·1%)	<10/24 174 (<0·1%)
Nephrotic syndrome	1031/1 267 685 (0·1%)	304/825 417 (<0·1%)	278/388 125 (0·1%)	128/222 067 (0·1%)	610/316 776 (0·2%)	26/16 259 (0·2%)	25/24 174 (0·1%)
IgA nephropathy	953/1 267 685 (0·1%)	310/825 417 (<0·1%)	330/388 125 (0·1%)	204/222 067 (0·1%)	313/316 776 (0·1%)	<10/16 259 (<0·1%)	31/24 174 (0·1%)
Focal segmental glomerulonephritis	82/1 267 685 (<0·1%)	20/825 417 (<0·1%)	18/388 125 (<0·1%)	<10/222 067 (<0·1%)	52/316 776 (<0·1%)	<10/16 259 (<0·1%)	<10/24 174 (<0·1%)
Minimum change glomerulopathy	45/1 267 685 (<0·1%)	10/825 417 (<0·1%)	<15/388 125 (<0·1%)	<10/222 067 (<0·1%)	<35/316 776 (<0·1%)	16/16 259 (0·1%)	<10/24 174 (<0·1%)

(Table continues on next page)

	All patients (N=2 564 729)	Patients prescribed inhaled corticosteroids (N=913 733)	Patients prescribed cutaneous corticosteroids (N=1 006 463)	Patients prescribed nasal corticosteroids (N=546 570)	Patients prescribed oral corticosteroids (N=440 775)	Patients prescribed ocular corticosteroids (N=47 249)	Patients prescribed rectal or cutaneous corticosteroids (N=95 971)
(Continued from previous page)							
Acute interstitial nephritis	311/1 267 685 (<0·1%)	82/825 417 (<0·1%)	84/388 125 (<0·1%)	36/222 067 (<0·1%)	169/316 776 (0·1%)	16/16 259 (0·1%)	<10/24 174 (<0·1%)
Post kidney transplant	1255/1 267 685 (<0·1%)	164/825 417 (<0·1%)	224/388 125 (0·1%)	136/222 067 (0·1%)	936/316 776 (0·3%)	27/16 259 (0·2%)	27/24 174 (0·1%)
Endocrine							
Subacute thyroiditis	809/1 267 685 (0·1%)	258/825 417 (<0·1%)	282/388 125 (0·1%)	195/222 067 (0·1%)	151/316 776 (<0·1%)	10/16 259 (0·1%)	20/24 174 (0·1%)
Hyperaldosteronism	128/1 267 685 (<0·1%)	42/825 417 (<0·1%)	39/388 125 (<0·1%)	27/222 067 (<0·1%)	37/316 776 (<0·1%)	<10/16 259 (<0·1%)	<10/24 174 (<0·1%)
Graves ophthalmopathy	631/1 267 685 (<0·1%)	241/825 417 (<0·1%)	210/388 125 (0·1%)	134/222 067 (0·1%)	141/316 776 (<0·1%)	34/16 259 (0·2%)	18/24 174 (0·1%)
Adrenal insufficiency	3509/1 267 685 (0·3%)	1109/825 417 (0·1%)	586/388 125 (0·2%)	455/222 067 (0·2%)	3375/316 776 (1·1%)	45/16 259 (0·3%)	50/24 174 (0·2%)
Congenital adrenal hyperplasia	265/1 267 685 (<0·1%)	32/825 417 (<0·1%)	29/388 125 (<0·1%)	14/222 067 (<0·1%)	233/316 776 (0·1%)	34/16 259 (0·2%)	<10/24 174 (<0·1%)
Hypopituitarism	1288/1 267 685 (0·1%)	183/825 417 (<0·1%)	166/388 125 (<0·1%)	114/222 067 (0·1%)	1213/316 776 (0·4%)	11/16 259 (0·1%)	17/24 174 (0·1%)
Dermatology							
Eczema	229 382/1 267 685 (18·1%)	53 914/825 417 (6·5%)	184 224/388 125 (47·5%)	27 440/222 067 (12·4%)	25 272/316 776 (8·0%)	2421/16 259 (14·9%)	4569/24 174 (18·9%)
Vitiligo	1077/1 267 685 (0·1%)	507/825 417 (0·1%)	539/388 125 (0·1%)	215/222 067 (0·1%)	198/316 776 (0·1%)	18/16 259 (0·1%)	23/24 174 (0·1%)
Stevens–Johnson's syndrome	112/1 267 685 (<0·1%)	33/825 417 (<0·1%)	46/388 125 (<0·1%)	11/222 067 (<0·1%)	46/316 776 (<0·1%)	11/16 259 (0·1%)	<10/24 174 (<0·1%)
Pyoderma gangrenosum	296/1 267 685 (<0·1%)	71/825 417 (<0·1%)	192/388 125 (<0·1%)	27/222 067 (<0·1%)	128/316 776 (<0·1%)	<10/16 259 (<0·1%)	<10/24 174 (<0·1%)
Lichen planus	9195/1 267 685 (0·7%)	2007/825 417 (0·2%)	6139/388 125 (1·6%)	1723/222 067 (0·8%)	2607/316 776 (0·8%)	141/16 259 (0·9%)	305/24 174 (1·3%)
Blistering conditions	3118/1 267 685 (0·2%)	734/825 417 (0·1%)	2104/388 125 (0·5%)	386/222 067 (0·2%)	1536/316 776 (0·5%)	93/16 259 (0·6%)	65/24 174 (0·3%)
Alopecia	26 620/1 267 685 (2·1%)	7245/825 417 (0·9%)	14 918/388 125 (3·8%)	5408/222 067 (2·4%)	3222/316 776 (1·0%)	296/16 259 (1·8%)	1005/24 174 (4·2%)
Gastrointestinal							
Inflammatory bowel disease	22 037/1 267 685 (1·7%)	6380/825 417 (0·8%)	7353/388 125 (1·9)	3852/222 067 (1·7)	8606/316 776 (2·7%)	341/16 259 (2·1%)	898/24 174 (3·7%)
Oral lichen planus	2364/1 267 685 (0·2%)	565/825 417 (0·1%)	1056/388 125 (0·3%)	598/222 067 (0·3%)	1211/316 776 (0·4%)	47/16 259 (0·3%)	89/24 174 (0·4%)
Eosinophilic oesophagitis	909/1 267 685 (0·1%)	474/825 417 (0·1%)	199/388 125 (0·1%)	158/222 067 (0·1%)	316/316 776 (0·1%)	<10/16 259 (<0·1%)	21/24 174 (0·1%)
Autoimmune hepatitis	1230/1 267 685 (0·1%)	222/825 417 (<0·1%)	232/388 125 (0·1%)	141/222 067 (0·1%)	975/316 776 (0·3%)	10/16 259 (0·1%)	31/24 174 (0·1%)
Neurology							
Myasthenia gravis	1441/1 267 685 (0·1%)	317/825 417 (<0·1%)	287/388 125 (0·1%)	186/222 067 (0·1%)	1109/316 776 (0·4%)	29/16 259 (0·2%)	43/24 174 (0·2%)
Multiple sclerosis	4760/1 267 685 (0·4%)	1383/825 417 (0·2%)	2141/388 125 (0·6%)	929/222 067 (0·4%)	974/316 776 (0·3%)	96/16 259 (0·6%)	192/24 174 (0·8%)
Chronic inflammatory demyelinating polyradiculoneuropathy	148/1 267 685 (<0·1%)	36/825 417	55/388 125 (<0·1%)	21/222 067 (<0·1%)	77/316 776 (<0·1%)	<10/16 259 (<0·1%)	<10/24 174 (<0·1%)
Bell's Palsy	4182/1 267 685 (0·3%)	1219/825 417 (0·1%)	1138/388 125 (0·3%)	666/222 067 (0·3%)	1913/316 776 (0·6%)	76/16 259 (0·5%)	118/24 174 (0·5%)
Ophthalmology							
Uveitis (non-infectious)	7480/1 267 685 (0·6%)	1908/825 417 (0·2%)	1898/388 125 (0·5%)	1124/222 067 (0·5%)	1573/316 776 (0·5%)	3461/16 259 (21·3%)	161/24 174 (0·7%)

Data are n (%) or n/N (%), corresponding to the absolute numbers of patients and the proportion of patients with each indication out of those with at least one indication for each column. Proportions are not mutually exclusive as multiple clinical indications could have been recorded. Numbers fewer than ten were supressed, as were any other data where counts of fewer than ten could be calculated.

***Table:* Clinical indications by route of administration**

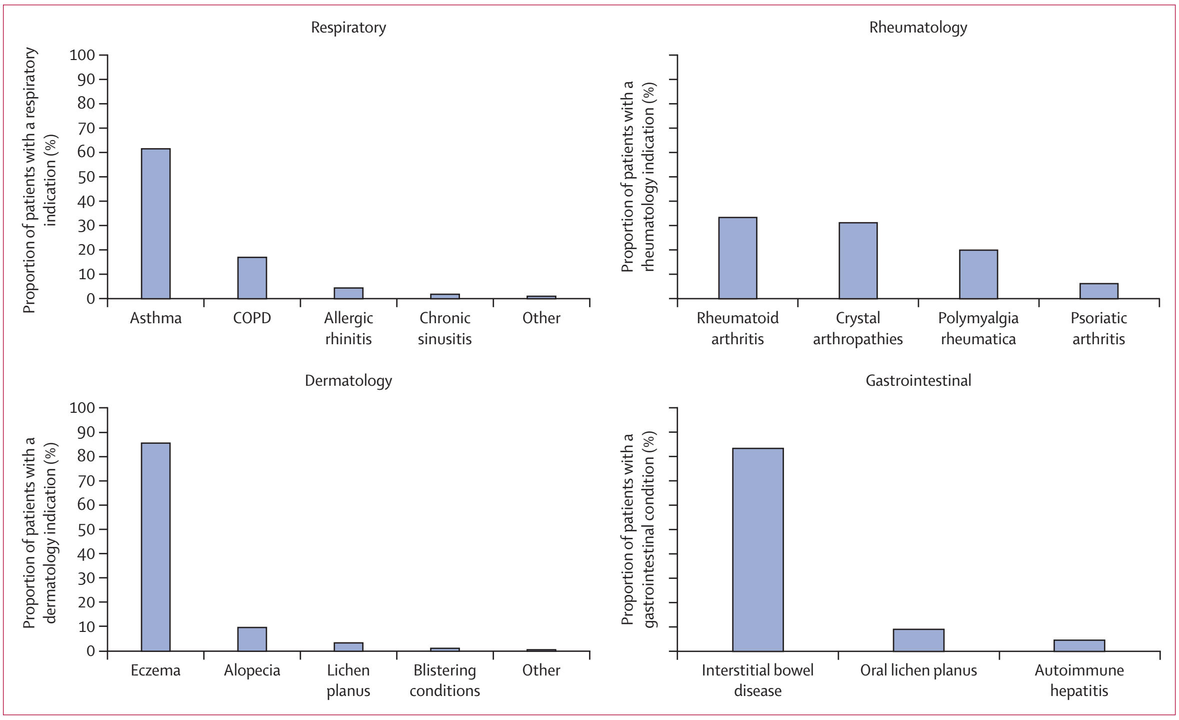

Figure 2: **Most common clinical indications for which corticosteroids were prescribed in the four most common disease systems**
Estimates report the proportion of people with at least one specified indication in each of the four most common disease systems. Further data are provided in the appendix (p 17). Proportions are not mutually exclusive because multiple clinical indications could have been recorded in the same disease system. COPD=chronic obstructive pulmonary disease.

Of those who had at least two clinical indications (N=265 633), 111 624 (42·0%) had at least two indications that fell within the same disease system. A total of 154 009 (58·0%) had at least two indications that fell within two different disease systems. The most common two disease systems that were recorded in combination within patients prescribed corticosteroids were respiratory and dermatology-related clinical indications, whereby 82 822 (53·8%) of patients with at least two indications in at least two different disease systems had indications for both respiratory disease and dermatology-related diseases (figure 3). The second most common combination was respiratory and rheumatology-related diseases, whereby 46 137 (30·0%) of patients had clinical indications for both types of diseases. Dermatology and rheumatology-related diseases were the third most common combination of disease indications with 11 445 (7·4%) of patients having clinical indications for both types of diseases (figure 3). Respiratory-related diseases were also commonly found in combination with renal, endocrine, gastrointestinal, and neurological diseases and gastrointestinal diseases were commonly found in combination with rheumatological and dermatological diseases (figure 3). The proportions for all combinations of disease systems are presented in the appendix (p 21).

Discussion

This large, population-based study of routinely collected primary care data in England showed that corticosteroids remain widely prescribed across adult patients. We found differences in corticosteroid prescribing patterns across route of administration, whereby inhaled corticosteroids were the most prescribed route and were more commonly prescribed for long durations of time than were other routes of corticosteroid administration. Additionally, clinical indications were infrequently recorded before corticosteroids were prescribed and differences in prevalence of clinical indications varied by route of administration. We found that respiratory and dermatology-related indications commonly co-occurred in patients prescribed corticosteroids, followed by respiratory and rheumatology-related indications.

Inhaled corticosteroids were the most prescribed administration route in this study, and were commonly prescribed for long durations, with nearly half of individuals receiving inhaled corticosteroids for the full year. This finding was probably driven by the high prevalence of chronic respiratory conditions in the UK. In 2019, the prevalence of asthma was approximately 11·5% and the prevalence of COPD was about 4·3% across the UK.[20] Chronic respiratory conditions often require continuous long-term treatment to maintain symptom control and prevent exacerbations, which explains the higher frequency and duration of inhaled corticosteroid prescriptions. Previous studies have also found that prevalence of inhaled corticosteroid treatment for asthma has increased over time in the UK.[21] People with respiratory disease are also often prescribed oral corticosteroids to treat exacerbations, which is why the proportion of oral corticosteroid use was relatively high in people with respiratory diseases, notably COPD.[22,23] Although we did not explore this in our study, our results demonstrate the importance of investigating the burden of corticosteroid use in people with respiratory diseases. Cutaneous corticosteroids were also commonly prescribed but were used for shorter durations, which might be due to the episodic nature of some dermatological conditions. For example, although the prevalence of eczema in adults is approximately 10%, eczema is a condition whereby symptoms can become worse for short periods of time and treatment for management of symptoms reflect these changes.[24] The episodic nature of skin conditions also explains why the proportion of unique corticosteroid prescriptions for cutaneous formulations was much lower than that for the proportion of patients who were prescribed at least one cutaneous corticosteroid, as steroids might be prescribed for acute instances.

Rheumatological conditions are less prevalent in the UK than respiratory and skin conditions, with a prevalence of 0·5–1·0% for rheumatoid arthritis and 0·8% for polymyalgia rheumatica.[25,26] For renal conditions, diseases such as nephrotic syndrome, glomerulonephritis, and interstitial nephritis are rarer and often managed in specialist care. Although the low prevalence of some of these conditions might explain the low proportion of corticosteroids prescribed for each disease system or route of administration, this might not always be the case. For example, various conditions might be more prevalent, but fewer patients require corticosteroids, and other conditions might be rare but result in high corticosteroid use.

Notably, only half of patients prescribed corticosteroids in 2023 had a clinical indication for corticosteroid use in the previous 2 years. Specifically, inhaled and oral corticosteroids were more likely to have a documented clinical indication than were other routes of administration, such as cutaneous, nasal, ocular, or rectal. This discrepancy might be driven by differences in disease-specific practices and progression. For example, patients with asthma or COPD have structured processes to review management of symptoms and disease progression, which can improve coding practices in these conditions. Despite this, the high proportion of corticosteroid prescriptions without an associated clinical indication raises concerns about the quality and completeness of clinical documentation in primary care. This lack of coding might hinder appropriate monitoring and review of corticosteroid use, making it difficult for clinicians to determine when corticosteroid treatment should be adjusted or discontinued. This is especially important for conditions with episodic or relapsing disease phenotypes—such as eczema, allergic rhinitis, or some musculoskeletal conditions—for which regular assessment is crucial to avoid unnecessary long-term exposure. Furthermore, a recording-lag from the COVID-19 pandemic might have led to low interaction of patients with primary care services, resulting in the under-reporting of clinical indications in our study.

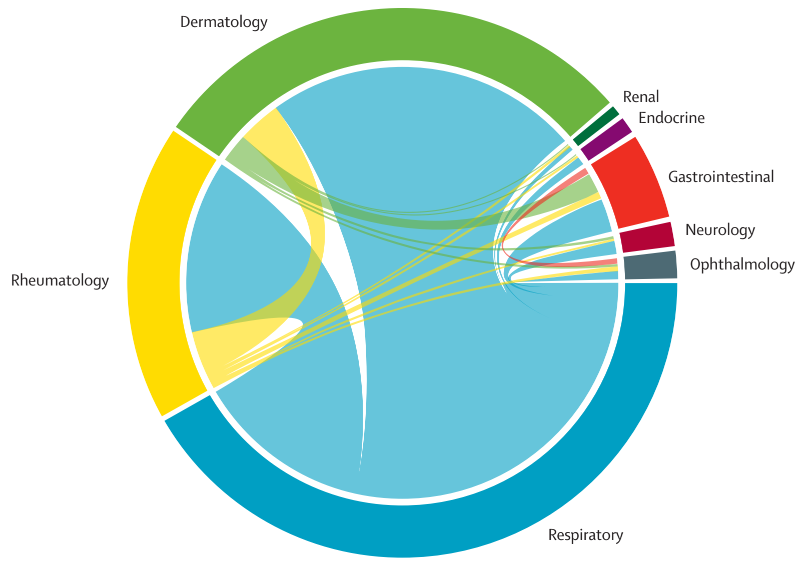

Figure 3: **Combinations of clinical indications in people with at least two clinical indications in two disease systems**

Chord diagram showing intersections between each disease system among those with at least one corticosteroid prescribed across at least two different disease systems. The outer arcs represent the proportion of corticosteroid use attributed to each disease system. The connecting chords illustrate where systems overlap, with chord width indicating the size of each overlap (ie, the proportions of patients who have each combination).

Where multiple clinical indications were recorded, the most common combination involved respiratory and dermatological conditions. The association between asthma and eczema is well-established in the literature. A Swedish study of people with asthma reported that 22% of patients also had a diagnosis of eczema.[27] Both conditions share common risk factors, including exposure to environmental allergens or pollution, genetic susceptibility, and overlapping inflammatory pathways.[28,29] Respiratory and rheumatological conditions were also frequently co-occurring, consistent with literature showing that COPD and rheumatoid arthritis are interlinked through mechanisms of systemic inflammation.[30] Overall, these findings highlight the complex interplay of corticosteroid prescribing in the context of multiorgan inflammatory disease.

Previous studies have reported the high burden of corticosteroid prescribing and associated harms, including adverse drug reactions and long-term effects such as immunosuppression, metabolic dysfunction, and the development of further conditions such as pneumonia in people with asthma.[6,8,10,31] The overuse and potential misuse of corticosteroids, particularly in the absence of clear clinical indications, is a concern. Our study supports these findings, highlighting a considerable proportion of corticosteroid prescriptions without a clinical indication in primary care. This sets the foundation for future research on corticosteroid-related adverse events, misuse, and discontinuation in primary care. Additionally, the high burden of corticosteroid prescribing observed in our study will probably involve considerable cost implications for the National Health Service. Given that the 2023 population of England was 57·7 million people, and assuming our results are generalisable to the wider English population, our results suggest that more than 9·2 million people were prescribed corticosteroids in 2023. Although some formulations, such as low-dose oral corticosteroids, are relatively inexpensive, others such as topical or specialist preparations are substantially more costly. Not only are the direct costs of corticosteroid prescriptions substantial, but the additional costs associated with managing their potential side-effects further increase the overall burden on the health-care system. Contextualising our findings in relation to prescribing guidelines for individual clinical indications is important. Certain indications result in a prescription of short course or one-off corticosteroids for flare-ups such as respiratory exacerbations, while other corticosteroids are prescribed long term to manage chronic conditions such as COPD. For example, we found that inhaled medications were more likely to be prescribed continuously than were other preparations, which might relate to the chronic nature of indications requiring inhalers. Future research should explore this to not only understand the burden of corticosteroids within indications but to determine adherence to prescribing guidelines.

To our knowledge, this is the first study to describe current prescribing patterns of corticosteroids across multiple conditions using routinely collected data of more than 2·5 million people in England. Although this study provides a high-level overview, there are several limitations. First, our analysis was restricted to primary care prescription data and therefore corticosteroids prescribed in hospital and over-the-counter medications were not included. As a result, the total burden of corticosteroid prescribing might have been underestimated; however, in terms of over-the-counter medications, very few corticosteroid medications are available, and usually at low dose for short periods of time, such as 1% hydrocortisone for mild skin conditions or 50 μg beclometasone nasal sprays for hay fever. Second, we cannot be certain that a specific corticosteroid prescription was prescribed due to a specific clinical indication as these are not recorded routinely when prescribing a drug in UK primary care, and we aim to develop methodological processes exploring this in the future. Third, the set of disease systems included in this study was defined a priori based on existing literature and clinical input. However, we acknowledge that some conditions, such as cancer and potentially non-specific clinical indications recorded alongside prescriptions, which would not have been included, might have led to an underestimation of the overall burden of corticosteroid prescribing. Despite this, our analysis captured the major disease systems in which corticosteroids are mostly prescribed in primary care, and we believe our estimates reflect the burden of corticosteroid use in this setting. Similarly, clinical indications were identified based on previous literature and clinical input; however, this might have led to the under-reporting of clinical indications seen in our study. Additionally, although we reported potency and dose of drug substances by route of administration, we did not investigate dose in more detail, such as cumulative dose, given the high-level nature of our study. However, this will be explored in further work. Similarly, future work will investigate the burden of corticosteroid use in relation to other medications for different conditions, which we were unable to do with our CPRD data. Furthermore, data were based on prescribed corticosteroids, not dispensed, and should be interpreted accordingly. Finally, although our study provides a high-level overview of corticosteroid use, it does not standardise prescribing patterns by disease system or clinical indication. As a result, it might overlook important differences such as rare conditions with a high corticosteroid use versus more prevalent diseases for which corticosteroid use is less common. We intend to explore these condition-specific patterns in greater detail in future analyses. Overall, our study results are generalisable to the wider population of primary care practices in England and should be interpreted as such.

To our knowledge, this is the first study to explore corticosteroid prescribing patterns across a range of conditions in England. Corticosteroids are widely prescribed across multiple routes of administration, with inhaled formulations the most common and frequently prescribed in the long term. The co-occurrence of conditions requiring corticosteroids also highlights the importance of optimal management and prescribing in patients with multi-organ disease. Given the increasing numbers of corticosteroid-sparing treatments such as biologics, further research evaluating their use in more detail, as well as safety and costs of corticosteroid use, is needed to determine the appropriateness of corticosteroids for specific conditions.

Contributors

HW analysed the data and wrote the original manuscript. MJ, FAP, SBe, and JKQ contributed to the creation of code lists and reviewed and edited the manuscript. CK, DH, SBr, ALN, JR, TS, SL, AS, and SBe reviewed and edited the manuscript. JKQ and HW accessed and verified the underlying data. All authors confirm they had full access to all the data in the study and accept responsibility for the decision to submit for publication.

Declaration of interests

FAP received an investigator-led grant from CSL-Vifor, outside of the submitted work. SBe received consulting fees from AstraZeneca and GSK, outside of the submitted work. AS and JKQ received grant funding from Health Data Research UK, which was used for the submitted work. JKQ also received grants from UK Research and Innovation, the National Institute for Health and Care Research, Health Data Research UK, Boehringer Ingelheim, AstraZeneca, and Insmed; and consulting fees from GSK and BMI, all outside of the submitted work. DH is a medical advisor to STERITAS. SBr received grants from Wellcome Trust, UK Research and Innovation, Medica Medical Research Council, Rosetrees Trust, Stoneygates Trust, British Skin Foundation, Charles Wolfson Charitable Trust, anonymous donations from people with eczema, Unilever, Pfizer, Abbvie, Sosei-Heptares, Janssen, European Lead Factory (multiple industry partners), and the BIOMAP consortium, all outside of the submitted work. MJ is funded by a National Institute for Health and Care Research (NIHR) Advanced Fellowship [NIHR301413]. The views expressed in this publication are those of the authors and not necessarily those of the NIHR, NHS, or the

UK Department of Health and Social Care. All other authors declare no competing interests.

Data sharing

Data may be obtained from a third party and are not publicly available. Datasets generated or analysed in this study are not publicly available; however, data are available on request from the Clinical Practice Research Datalink (CPRD; https://www.cprd.com/). Their provision requires the purchase of a license from the CPRD, and this license does not permit the authors to make them publicly available to all. This work used data from the version of CPRD Aurum collected in March, 2025, and the data selected is clearly specified in the Methods section. To allow identical data to be obtained by others via the purchase of a license, the code lists will be provided upon request to the corresponding author.

References

1 Coutinho AE, Chapman KE. The anti-inflammatory and immunosuppressive effects of glucocorticoids, recent developments and mechanistic insights. *Mol Cell Endocrinol* 2011; **335:** 2–13.
2 Zoorob RJ, Cender D. A different look at corticosteroids. *Am Fam Physician* 1998; **58:** 443–50.
3 Alangari AA. Corticosteroids in the treatment of acute asthma 2014; **9:** 187–92.
4 Quint JK, Ariel A, Barnes PJ. Rational use of inhaled corticosteroids for the treatment of COPD. *NPJ Prim Care Respir Med* 2023; **33:** 27.
5 NICE. Scenario: corticosteroids. 2025. https://cks.nice.org.uk/topics/corticosteroids- oral/management/corticosteroids/ (accessed Dec 2, 2025).
6 Paradkar S. Reported adverse drug reactions during the use of corticosteroids in a tertiary care hospital. *Ther Innov Regul Sci* 2019; **53:** 128–31.
7 Barnes PJ. Corticosteroid resistance in patients with asthma and chronic obstructive pulmonary disease. *J Allergy Clin Immunol* 2013; **131:** 636–45.
8 Barnes PJ, Adcock IM. Glucocorticoid resistance in inflammatory diseases. *Lancet* 2009; **373:** 1905–17.
9 Mehta AB, Nadkarni NJ, Patil SP, et al. Topical corticosteroids in dermatology. *Indian J Dermatol Venereol Leprol* 2016; **82:** 371–78.
10 Heffler E, Madeira LNG, Ferrando M, et al. Inhaled corticosteroids safety and adverse effects in patients with asthma. *J Allergy Clin Immunol Pract* 2018; **6:** 776–81.
11 Medicines and Healthcare products Regulatory Agency. Topical steroid withdrawal reactions: a review of the evidence. 2021. https://www.gov.uk/government/publications/topical-steroid-withdrawal-reactions-a-review-of-the-evidence (accessed Dec 2, 2025).
12 Wallace BA-O, Waljee AK, Weissman A, Caverly TJ, Saini SD. Overuse of glucocorticoids in rheumatoid arthritis: a national survey of primary care physicians. *Rheumatol Adv Pract* 2018; **23** (3): rky049.
13 Wolf A, Dedman D, Campbell J, et al. Data resource profile: Clinical Practice Research Datalink (CPRD) Aurum. *Int J Epidemiol* 2019; **48:** 1740–1740g.
14 Mahadevan P, Harley M, Fordyce S, et al. Completeness and representativeness of small area socioeconomic data linked with the UK Clinical Practice Research Datalink (CPRD). *J Epidemiol Community Health* 2022; **76:** 880–86.
15 Shiekh SI, Harley M, Ghosh RE, et al. Completeness, agreement, and representativeness of ethnicity recording in the United Kingdom's Clinical Practice Research Datalink (CPRD) and linked Hospital Episode Statistics (HES). *Popul Health Metr* 2023; **21:** 3.
16 Graul E, Stone P, Massen G, et al. Determining prescriptions in electronic healthcare record data: methods for development of standardized, reproducible drug codelists. *JAMIA Open* 2023; **6:** ooad078.
17 Moore E, Al Sallakh M, Olajide O, et al. Mapping use of high dose or long-term oral glucocorticoids and steroid-sparing strategies in adults with chronic conditions: a rapid scoping review of reviews. *Pharmacoepidemiol Drug Saf* 2025; **34:** e70233.
18 Ramirez Medina CR, Lyon M, Davies E, et al. Clinical indications associated with new opioid use for pain management in the United Kingdom: using national primary care data. *Pain* 2025; **166:** 656–66.
19 NHS England. Output your results. https://digital.nhs.uk/services/secure-data-environment- service/access-and-support-for-the-secure-data-environment/user-guides/output-your-results (accessed Dec 2, 2024).
20 Whittaker H, Kramer Fiala Machado A, Hatam S, et al. Incidence and prevalence of asthma, chronic obstructive pulmonary disease and interstitial lung disease between 2004 and 2023: harmonised analyses of longitudinal cohorts across England, Wales, South-East Scotland and Northern Ireland. *Thorax* 2025; **80:** 466–77.
21 Bloom CI, Saglani S, Feary J, et al. Changing prevalence of current asthma and inhaled corticosteroid treatment in the UK: population-based cohort 2006–2016. *Eur Respir J* 2019; **53:** 1802130.
22 Tran TN, King E, Sarkar R, et al. Oral corticosteroid prescription patterns for asthma in France, Germany, Italy and the UK. *Eur Respir J* 2020; **55:** 1902363.
23 Heatley H, Tran TN, Bourdin A, et al. Observational UK cohort study to describe intermittent oral corticosteroid prescribing patterns and their association with adverse outcomes in asthma. *Thorax* 2023; **78:** 860–67.
24 Cork MJ, Danby SG, Ogg GS. Atopic dermatitis epidemiology and unmet need in the United Kingdom. *J Dermatolog Treat* 2020; **31:** 801–09.
25 Aletaha D, Smolen JS. Diagnosis and management of rheumatoid arthritis: a review. *JAMA* 2018; **320:** 1360–72.
26 Partington R, Muller S, Helliwell T, et al. Incidence, prevalence and treatment burden of polymyalgia rheumatica in the UK over two decades: a population-based study. *Ann Rheum Dis* 2018; **77:** 1750–56.
27 Pullerits T, Rönmark EP, Ekerljung L, et al. The triad of current asthma, rhinitis and eczema is uncommon among adults: Prevalence, sensitization profiles, and risk factors. *Respir Med* 2021; **176:** 106250.
28 Yaneva MA-O, Darlenski R. The link between atopic dermatitis and asthma- immunological imbalance and beyond. *Asthma Res Pract* 2021; **7:** 16.
29 Gu X, Li Z, Su J. Air pollution and skin diseases: a comprehensive evaluation of the associated mechanism. *Ecotoxicol Environ Saf* 2024; **278:** 116429.
30 Yang K, Wang L, Chen S, Chen R. Longitudinal reciprocal association between rheumatoid arthritis and chronic obstructive pulmonary disease and mediation of inflammation. *Rheumatology (Oxford)* 2024; **63:** 2763–69.
31 von Bülow A, Hansen S, Sandin P, et al. Use of high-dose inhaled corticosteroids and risk of corticosteroid-related adverse events in asthma findings from the NORDSTAR cohort. *J Allergy Clin Immunol Pract* 2025; **13:** 1609–19.e5.